INDEPENDENCE OF THE FUNCTOR RF_! FROM THE CHOSEN COMPACTIFICATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-etale.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Editorial note. Disclosure: two soft spots. (1) The reduction from the compatibility of compositions to the agreement of the underlying structure maps is introduced by 'A calculation shows that it suffices to prove' (proof line 3361); the calculation itself is not displayed – it is the definitional unfolding of the two composites of functor isomorphisms. (2) The final agreement 'follows immediately from Lemma lemma-shriek-proper-and-open-compose' (proof lines 3369-3372), a named tool; the special-case identification of the map of lemma-shriek-lqf-and-proper with the map of lemma-shriek-proper-and-open is stated via the separatedness of the j_i. The cofiltered cancellation logic is spelled out in the footnote at proof lines 3345-3348. No verbatim 'we omit' occurs in the proof.

Allowed standard prerequisites (editorial).

- Cofiltered category of compactifications (More on Flatness, Theorem flat-theorem-nagata 0F41 with Lemmas flat-lemma-compactifications-cofiltered 0ATU and flat-lemma-compactifyable 0A9Z). cited at proof lines 3313-3316. Provides the cofiltered category of compactifications within which the functors are compared, and the composition criterion for functorial isomorphisms over it.
- Canonical maps from lower shriek to derived pushforward (Lemma lemma-shriek-lqf-and-proper, canonical tag 0F7G). cited at proof lines 3339, 3364-3366 and 3369. Supplies the structure maps j_! into derived pushforwards from which the isomorphisms attached to morphisms of compactifications are built.
- Composition compatibility of the exchange maps (Lemma lemma-shriek-proper-and-open-compose, canonical tag 0F7C). cited at proof lines 3369-3372. The terminal tool: the composition of structure maps agrees with the structure map of the composite morphism of compactifications.

Source reference. Original lemma 0ATU: sources/stacks-chow/tags.txt, source lines 13627–13627. Cited at proof lines 3313-3316 and 3342-3343: the category of compactifications of X over Y is cofiltered, used for the existence of morphisms and the composition criterion.

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Source reference. Original lemma 0A9Z: sources/stacks-chow/tags.txt, source lines 12967–12967. Cited at proof lines 3313-3316: compactifiability underlying the category of compactifications of X over Y.

Source reference. Original theorem 0F41: sources/stacks-chow/tags.txt, source lines 18886–18886. Cited at proof lines 3313-3314: existence of compactifications anchoring the cofiltered category.

Source reference. Original lemma 0F7G: sources/stacks-chow/tags.txt, source lines 19006–19006. Cited at proof lines 3339, 3364-3366 and 3369: the structure maps j_! into derived pushforwards from which the isomorphisms between the functors of different compactifications are built.

Source reference. Original lemma 0F7B: sources/stacks-chow/tags.txt, source lines 19001–19001. Cited at proof lines 3369-3371: the map of lemma-shriek-lqf-and-proper is a special case of its map since the compactification maps j_i are separated.

Source reference. Original lemma 0F7C: sources/stacks-chow/tags.txt, source lines 19002–19002. Cited at proof lines 3371-3372: the composition of the structure maps agrees with the structure map of the composite morphism of compactifications, finishing the proof.

Lemma 0.2. *Let $f : X \to Y$ be a finite type separated morphism of quasi-compact and quasi-separated schemes. The functors $Rf_!$ constructed above are, up to canonical isomorphism, independent of the choice of the compactification.*

Proof. We will prove this for the functor $Rf_! : D(X_{étale}, \Lambda) \to D(Y_{étale}, \Lambda)$ when Λ is a torsion ring; the case of the functor $Rf_! : D^+_{tors}(X_{étale}, \Lambda) \to D^+_{tors}(Y_{étale}, \Lambda)$ is proved in exactly the same way.

Consider the category of compactifications of X over Y, which is cofiltered according to More on Flatness, Theorem 0F41 and Lemmas 0ATU and 0A9Z. To every choice of a compactification

$$j : X \to \overline{X}, \quad \overline{f} : \overline{X} \to Y$$

the construction above associates the functor $R\overline{f}_* \circ j_! : D(X_{étale}, \Lambda) \to D(Y_{étale}, \Lambda)$. Let's be a little more explicit. Given a complex $\mathcal{K}^\bullet$ of sheaves of Λ-modules on $X_{étale}$, we choose a quasi-isomorphism $j_!\mathcal{K}^\bullet \to \mathcal{I}^\bullet$ into a K-injective complex of sheaves of Λ-modules on $\overline{X}_{étale}$. Then our functor sends $\mathcal{K}^\bullet$ to $\overline{f}_*\mathcal{I}^\bullet$.

Suppose given a morphism $g : \overline{X}_1 \to \overline{X}_2$ between compactifications $j_i : X \to \overline{X}_i$ over Y. Then we get an isomorphism

$$R\overline{f}_{2,*} \circ j_{2,!} = R\overline{f}_{2,*} \circ Rg_* \circ j_{1,!} = R\overline{f}_{1,*} \circ j_{1,!}$$

using Lemma 0F7G in the first equality.

To finish the proof, since the category of compactifications of X over Y is cofiltered, it suffices to show compositions of morphisms of compactifications of X over Y are turned into compositions of isomorphisms of functors[1]. To do this, suppose that $j_3 : X \to \overline{X}_3$ is a third compactification and that $h : \overline{X}_2 \to \overline{X}_3$ is a morphism of compactifications. Then we have to show that the composition

$$R\overline{f}_{3,*} \circ j_{3,!} = R\overline{f}_{3,*} \circ Rh_* \circ j_{2,!} = R\overline{f}_{2,*} \circ j_{2,!} = R\overline{f}_{2,*} \circ Rg_* \circ j_{1,!} = R\overline{f}_{1,*} \circ j_{1,!}$$

[1]Namely, if $\alpha, \beta : F \to G$ are morphisms of functors and $\gamma : G \to H$ is an isomorphism of functors such that $\gamma \circ \alpha = \gamma \circ \beta$, then we conclude $\alpha = \beta$.

is equal to the isomorphism of functors constructed using simply j_3, $g \circ h$, and j_1. A calculation shows that it suffices to prove that the composition of the maps

$$j_{3,!} \to Rh_* \circ j_{2,!} \to Rh_* \circ Rg_* \circ j_{1,!}$$

of Lemma 0F7G agrees with the corresponding map $j_{3,!} \to R(h \circ g)_* \circ j_{1,!}$ via the identification $R(h \circ g)_* = Rh_* \circ Rg_*$. Since the map of Lemma 0F7G is a special case of the map of Lemma 0F7B (as j_1 and j_2 are separated) this follows immediately from Lemma 0F7C. □